COURSE TITLE : YARN FORMATION I
COURSE CODE : 3064
COURSE CATEGORY : B
PERIODS/WEEK : 4
PERIODS/SEMESTER : 60
CREDITS : 4

TIME SCHEDULE

Module	Topics	Periods
1	Cotton Selection & Mixing	13
2	Blow room	17
3	Carding	17
4	Modern developments in Carding and card calculations	13
TOTAL		60

COURSE OUTCOME

Module	GO	Student will be able to
1	1	Understand ginning process, mixing , blending & mixing plans for various counts
	2	Know the working of various mixing machineries
2	1	Understand the working of various machineries in blow room & production, cleaning efficiency
	2	Know the working of scutcher & chute feed systems .
3	1	Understand the working and parts of Revolving Flat Carding Machine
	2	Know the maintenance operation, speeds, waste in a card and tandem carding
4	1	Understand the modern developments in carding
	2	Know the calculations in carding

SPECIFIC OUTCOME

MODULE I COTTON SELECTION & MIXING

1.1.0 Understand ginning process, mixing, blending & mixing plans for various counts
1.1.0 Define ginning and state the Objectives
1.1.1 List the types of gins for different cottons
1.1.2 State the Objectives of mixing different varieties of cotton fibres
1.1.3 List the mixing plans for coarse, medium and fine counts
1.1.4 Distinguish different methods of mixing

1.2.0 Know the working of various mixing machineries
1.2.1 Explain the working of Mixing bale opener

1.2.2 Explain the working of Multi mixer
1.2.3 Describe the working of Auto mixer
1.2.4 Explain the working of Blendomat

MODULE II BLOW ROOM

2.1.0 Understand the working of various machineries in blow room & production, cleaning efficiency
2.1.1 List the different contaminants in cotton fibre
2.1.2 State the need for opening and cleaning of fibres
2.1.3 Define cleaning efficiency
2.1.4 Classify Blow room lines and cleaning points
2.1.5 Explain the working of Hopper bale breaker
2.1.6 Describe the working of Mono Cylinder
2.1.7 Explain the working of ERM cleaner
2.1.8 Describe the working of Axi-flo cleaner
2.1.9 Determine the production and efficiency of blow room
2.1.10 Estimate the cleaning efficiency of the blow room

2.2.0 Know the working of scutcher & chute feed systems .
2.2.1 State the Objectives of Scutching
2.2.2 Illustrate the material passage through a Single process Scutcher
2.2.3 Describe the working of Hopper feeder
2.2.4 Illustrate the working of Bladed beaters
2.2.5 Describe the working of Krischner's beater
2.2.6 Explain the working of Lap Forming unit
2.2.7 Describe the working of Piano feed regulating mechanism
2.2.8 State the Objectives of Cages, Lap measuring motion, Auto doffing, Distributor
2.2.9 State the functions of Dust filter bag, magnetic traps
2.2.10 List various lap defects - their causes and remedies
2.2.11 Define chute feed system and describe the working

MODULE III CARDING

3.1.0 Understand the working and parts of Revolving Flat Carding Machine
3.1.1 List the Objectives of Carding
3.1.2 State the principle of Stripping and Carding action
3.1.3 Describe the passage of material in a Revolving Flat Carding Machine
3.1.4 State the functions of each part of Carding Machine
3.1.5 State the functions of Heal and Toe arrangement
3.1.6 Explain the working of H & B Flexible Bends
3.2.0 Know the maintenance operation, speeds, waste in a card and tandem carding
3.2.1 List and explain the maintenance operations given to cards
3.2.2 State the speeds of important organs of the cards
3.2.3 List the different types of waste in Carding
3.2.4 Define Tandem carding
3.2.5 List the defects in Carding and state their causes and remedies
3.2.6 Define Trailing and leading hooks in card sliver

MODULE IV MODERN DEVELOPMENTS IN CARDING AND CARD CALCULATIONS

4.1.0 Understand the modern developments in carding

4.1.1 List the features of LC 363/LC 361 high production Card
4.1.2 Identify the developments in Feed part and Licker– in region
4.1.3 Identify the developments in Carding region, Condensing region
4.1.4 State the need of auto levelers in cards
4.1.5 Illustrate the Closed loop & Open loop auto leveling system
4.1.6 Explain the working of Auto levelers employed in a card
4.1.7 State the features of Automatic waste extracting system
4.1.8 State the salient features of automatic doffing
4.1.9 List the various stop motions in a card

4.2.0 Know the calculations in carding

4.2.1 Define actual draft and mechanical draft
4.2.2 State the relation between actual draft and mechanical draft in card
4.2.3 Determine the Draft constant, draft and tension draft in card
4.2.4 Find the actual production of the Card
4.2.5 Estimate the efficiency of the Card
4.2.6 Determine the lap exhausting time in a card
4.2.7 Find the can filling time in a card

CONTENT DETAILS

MODULE I

COTTON SELECTION & MIXING

Define ginning- state the objectives ginning-list the types of gins for different cottons-State the objectives of mixing different varieties of cotton fibres- List the mixing plans for coarse, medium and fine counts-Distinguish different methods of mixing-Explain the working of Mixing bale opener-Expain the working of Multi mixer-Describe the working of Auto mixer-Explain the working of Blendomat.

MODULE II

BLOW ROOM

List the different contaminants in cotton fibre-State the need for opening and cleaning of fibres- Define cleaning efficiency-Classify Blow room lines and cleaning points-Explain the working of Hopper bale breaker-Describe the working of Mono Cylinder-Explain the working of ERM cleaner-Describe the working of Axi-flo cleaner-Determine the production and efficiency of blow room-Estimate the cleaning efficiency of the blow room-State the objectives of Scutching-Illustrate the material passage through a Single process Scutcher-Describe the working of Hopper feeder-Illustrate the working of Bladed beater 2 BB and 3 BB-Describe the working of Krischner's beater- Explain the working of Lap Forming unit-Describe the working of Piano feed regulating mechanism-State the objectives of Cages, Lap measuring motion, Auto doffing, 2 way Distributor-State the functions of Dust filter bag, magnetic traps-List various lap defects - their causes and remedies-Define chute feed system and describe the working.

MODULE III
CARDING

Define carding-List the Objectives of Carding- State the principle of Stripping and Carding action-Describe the passage of material in a Revolving Flat Carding Machine- State the functions of each part of Carding Machine-State the functions of Heal and Toe arrangement- Explain the working of H & B Flexible Bend-List and explain the maintenance operations given to cards-State the speeds of important organs of the cards such as cylinder, licker in, doffer and flats-List the different types of waste in Carding(Flat strips, Licker in droppings, cylinder & doffer flies, fan waster and roller waste-Define Tandem carding-List the defects in Carding and state their causes and remedies- Define Trailing and leading hooks in card sliver.

MODULE IV
MODERN DEVELOPMENTS IN CARDING AND CARD CALCULATIONS

List the features of LC 363/LC 361 high production Card-Identify the developments in Feed part and Licker in region such as chute feed system, improved feed roller, high speed licker in and 2 or 3 licker in and improved under grids -Identify the developments in Carding region such as pre & post carding segments, Condensing region-State the need of auto levelers in cards-Illustrate the Closed loop & Open loop auto leveling system-Explain the working of Auto levelers employed in a card such as short term, medium term and long term auto levelers-State the features of Automatic waste extracting system-State the salient features of automatic doffing-List the various stop motions in a card-Define actual draft and mechanical draft-State the relation between actual draft and mechanical draft in card-Determine the Draft constant, draft and tension (web & sliver) draft in card-Find the actual production of the Card-Estimate the efficiency of the Card-Determine the lap exhausting time in a card-Find the can filling time in a card.

REFERENCE BOOKS

1. W.S.Taggart	Cotton Spinning -	Mahajan Publishers
2. G.R.Merill	Cotton opening and picking	
3. G.R.Merill	Cotton carding	
4. Butter Worth	Manual of cotton spinning Volume I and II	
5. Venkita Subramani	Spun Yarn Technology Volume I and II	
6. Textile Association	Cotton Spinning	
7. Pattabiraman	Essential elements of Cotton Spinning	
8. W.S Taggart	Cotton Spinning Calculations	